

PORTFOLIO

Muhammad Rizqi

Software Engineer with 4+ years building scalable mobile, backend, and web applications across logistics, ERP, transportation, and enterprise systems.

Software Engineer

Mobile Developer

Full-Stack Developer



OPEN TO
COLLABORATE

EXPERIENCE

4+

years shipping production software

CURRENTLY

Technical Consultant (Mobile)

PT. Altitude Solutions · Dec 2025 – Present

Profile

Software Engineer with 4+ years of experience developing scalable mobile, backend, and web applications across logistics, ERP, transportation, and enterprise systems.

Experienced in Flutter, Android Native, Spring Boot Microservices, Next.js, OCR, AI-powered automation, and cloud-based services such as Firebase. Passionate about building maintainable software architectures, improving system performance, and delivering high-quality products in Agile environments.

Indonesian (native)

English (intermediate)

EDUCATION

BSc Information Engineering

Gadjah Mada University

August 2021 – October 2025

3.61

GPA out of 4.00

154

credits completed

Skills & tools

Six practice areas

LANGUAGES

Java Kotlin Dart JavaScript
TypeScript Python SQL PHP
HTML CSS

MOBILE

Flutter Android SDK
Jetpack Compose MVVM
Clean Architecture BLoC Cubit
GetIt Room Hive
Material Design

BACKEND

Spring Boot Node.js
Microservices REST API
GraphQL JWT Auth PostgreSQL
MySQL Firebase Supabase

FRONTEND

Next.js Nuxt.js Tailwind CSS
Bootstrap

AI & AUTOMATION

OCR OpenAI API
Prompt Engineering
Telegram Bot API
Document Processing

DEVOPS & TOOLS

Git GitHub GitLab Fastlane
Crashlytics Analytics Figma
Postman Android Studio
VS Code

Where I've worked

2023 – present

Technical Consultant (Mobile)

PT. Altitude Solutions · Dec 2025 – Present

Flutter driver app for a logistics ecosystem, Spring Boot microservices, ERP modules, and OCR-based invoice automation.

Website Administrator

Innovative Academy UGM · Aug – Dec 2025

Redesigned the official site, managed content and publication workflows, and supported innovation events.

Android Developer

Aino Indonesia · Feb – Apr 2025

Project Acasia, the Android ticketing system on TransJakarta buses, with Room migration and device utility features.

Mobile Developer

PT. Divistant Teknologi Indonesia · Sep 2023 – Dec 2024

Angkasa Pura flight app in Flutter, Clean Architecture and GraphQL, plus Jenkins and Fastlane CI/CD pipelines.

Android Developer Assistant

DTETI UGM · Feb – Aug 2023

IoT-based concrete cutting depth analysis for toll road infrastructure, synchronised over WebSocket.

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THESIS RESEARCH

A UWB-based Android application for high-precision real-time indoor positioning and self-localization.

TEACHING ASSISTANT

Single and Multivariable Calculus · Basic Programming Lab

Selected work

01	ARCIBO: Augmented Reality Civil Block	Mobile · 2024
02	MAHEZZA: Mental Health Puzzle Game	Mobile · 2023
03	Toko Management System	Mobile · 2022
04	Paragraph: Short Read, Easy Write	Mobile · 2021
05	Purwoasri Village Government Website	Web · WordPress



ARCIBO: Augmented Reality Civil Block

PROBLEM

Traditional houses are vital symbols of regional identity, yet they face development stagnation and pressure from modernization. Their knockdown Austronesian construction is rarely taught and rarely seen up close.

A knockdown toy paired with an AR mobile app recreates those houses as a preservation initiative, accessible to children, educators, and collectors.

1,340

ethnic groups

2,500

regional
languages

175

houses
documented,
2020



GOAL

Showcase traditional houses, educate people about their cultural significance, and promote active preservation.



How I built it

1

Cross-platform core

Built in Flutter for fast iteration and a light footprint, with BLoC, Cubit and GetIt handling state and dependency injection.

2

Layered architecture

Data, domain and presentation layers under MVVM. The data layer owns Firebase CRUD, the domain layer owns business logic.

3

Team lead & design

Led the team as CTEO through the PKM-K national competition and owned UI/UX end to end, down to the logo and brand graphics.

STACK

Flutter

Dart

Firebase

BLoC

Cubit

GetIt

Room

Camera API



What shipped

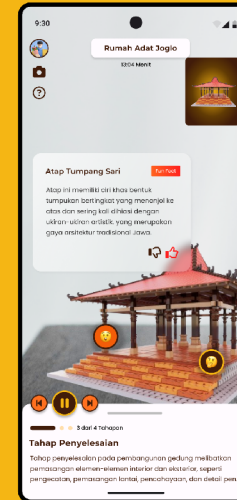
- 3D augmented reality models of traditional houses
- Educational descriptions, quizzes and fun facts
- Homepage with activity history and recent updates
- Gallery of completed games
- Progress tracking and achievement celebrations

LINKS

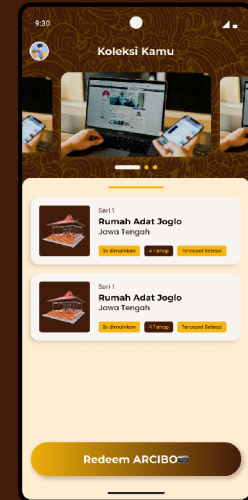
play.google.com/store/apps/details?id=com.arcibo.arcibo

github.com/m-rizqi/arcibo-app

Augmented Reality



Koleksi ARCIBO





MAHEZZA: Mental Health Puzzle Game

PROBLEM

Mental health among children and adolescents is a critical issue in Indonesia, and many parents lack both the knowledge and the time to properly support their children's well-being.

A knockdown puzzle board paired with a native Android app turns that gap into shared time: parent-guided storytelling, timers, calming music, and photo creation.

6.1%

adolescents
with mental
disorders, 2018

34.9%

reporting
mental health
issues, 2022

20%

of the population
is this generation



GOAL

Enrich the puzzle board with interactive features and educate children about mental health through parent-guided storytelling.



How I built it

1

Native for hardware

Kotlin with Jetpack Compose, chosen for easier access to phone functionality like audio playback and camera features.

2

Firebase backbone

Authentication, Firestore, Storage, Analytics and Crashlytics alongside a Room database and camera APIs.

3

CTO & designer

Ran the technical architecture for a year-long PKM-K project and owned UI/UX, banner, logo and brand graphics.

STACK

Kotlin

Jetpack Compose

Firebase

MVVM

Room

Camera API



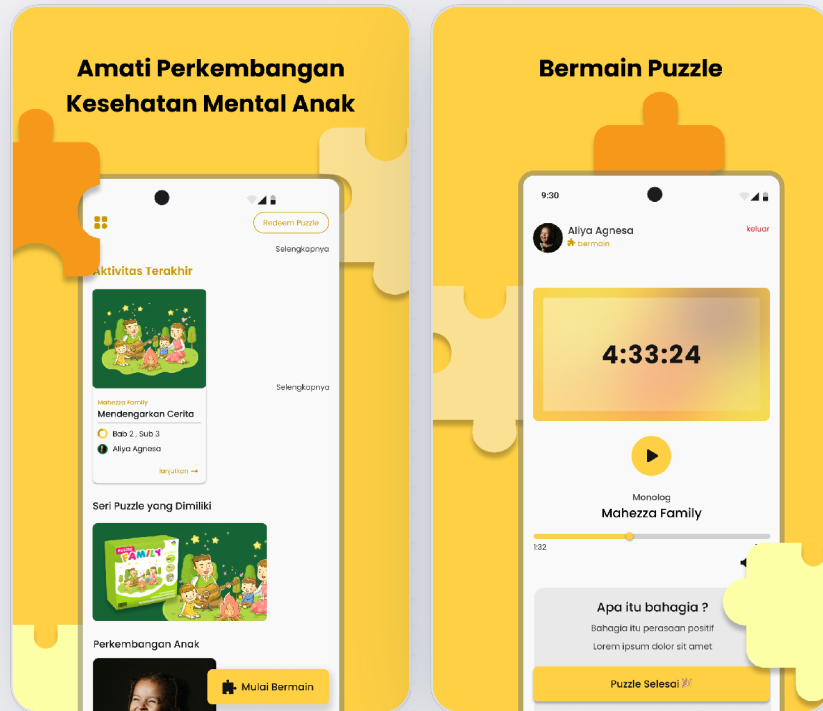
What shipped

- Interactive stories and challenges for parents to share with their children
- Timer and background music during gameplay
- Photo capture and twibbon creation to celebrate puzzle completion
- Homepage displaying recent activities

LINKS

play.google.com/store/apps/details?id=com.mahezza.mahezza

github.com/m-rizqi/mahezza





Toko Management System

PROBLEM

My parents run a store with no cashier technology: no computer, barcode scanner, or POS system. When they stepped away I had no way to know a product's price and had to keep asking them.

The constraint: store prices reliably using nothing but a phone, with fast lookup, and handle the store's two-tier pricing, since resellers pay a different price than regular consumers.

2

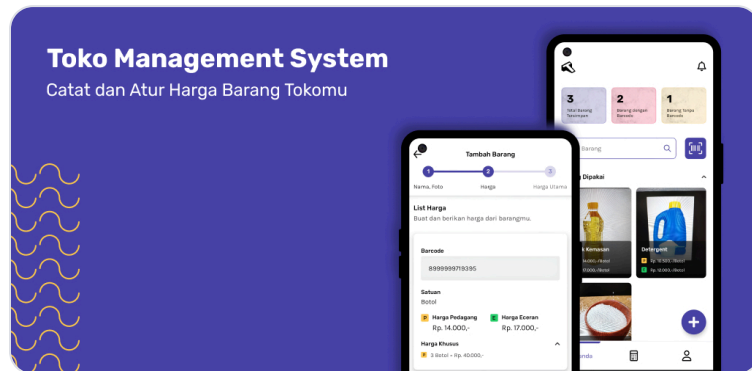
price tiers
per item

4mo

solo build, Sep –
Dec 2022

1

phone, no extra
hardware



GOAL

Cut the time sellers spend looking up prices, so they can serve more buyers without memorising the catalogue.



How I built it

1

Local-first data

Room stores everything on-device first, then syncs to Firebase Realtime Database and Storage on a user-set schedule, with a restore prompt at next login.

2

Phone as scanner

Google Vision turns the camera into a barcode scanner, with manual search covering items that have no barcode at all.

3

Kotlin + MVVM

Android View with Data Binding under MVVM, plus a Bluetooth printer library for receipts at the cashier step.

STACK

Kotlin

Android View

Firebase

Room

MVVM

Camera API



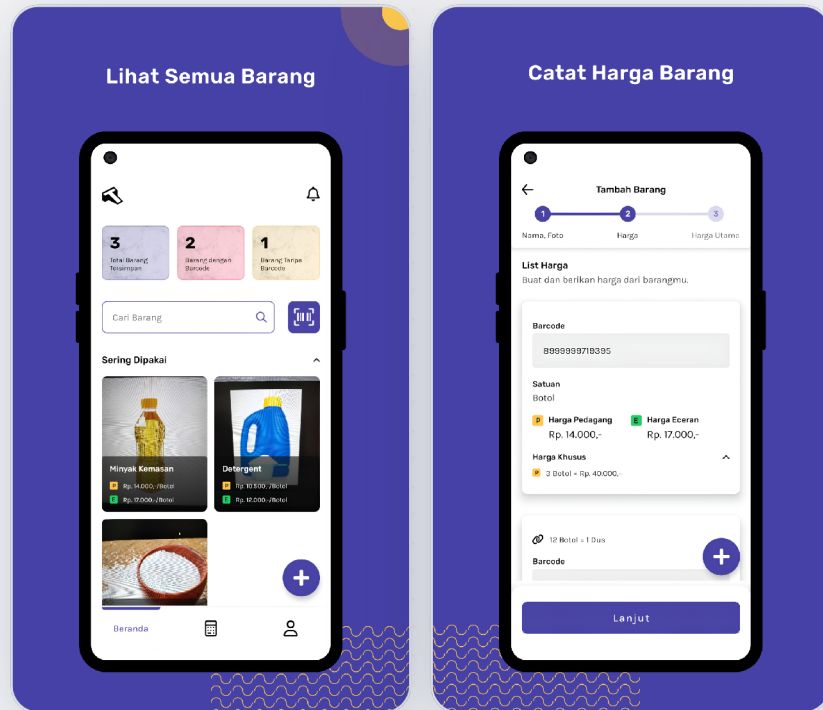
What shipped

- Search and add items by keyword or by barcode scan
- Separate consumer and retailer prices per item
- Different price per unit, with notes or special pricing for bulk quantities
- Frequently used and no-barcode items surfaced on the main page
- Cashier flow with Bluetooth receipt printing

LINKS

play.google.com/store/apps/details?id=com.rizqi.tms

github.com/m-rizqi/tms-app



Paragraph: Short Read, Easy Write

PROBLEM

Articles running into the thousands of words make people give up on reading before they even start, even when they still want the knowledge inside.

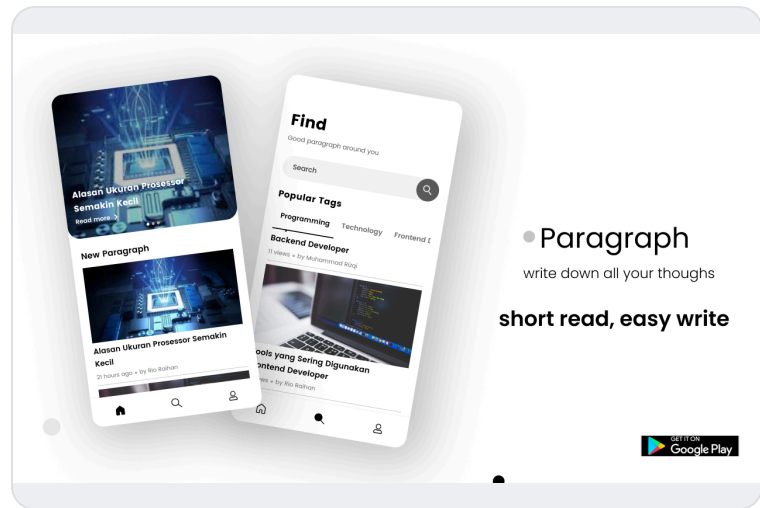
At the same time, some writers just want their work read by even a small audience. Paragraph tackles both by only ever surfacing short-form articles.

2mo

solo build, Oct – Nov 2021

3

roles: PM, designer,
developer



GOAL

Increase people's interest in reading by providing short but powerful readings.

How I built it

1

Design first

Designed the whole UI in Figma, then implemented it in Android Studio with XML views wired through View Binding.

2

Firebase CRUD

Authentication and Realtime Database behind posts, trending feeds, popular tags and multi-field search.

3

Swipeable feeds

ViewPager powers navigation between trending, new and tag-filtered feeds, with a full composer for writing.

STACK

Kotlin

Android View

View Binding

Firebase

Realtime Database

ViewPager

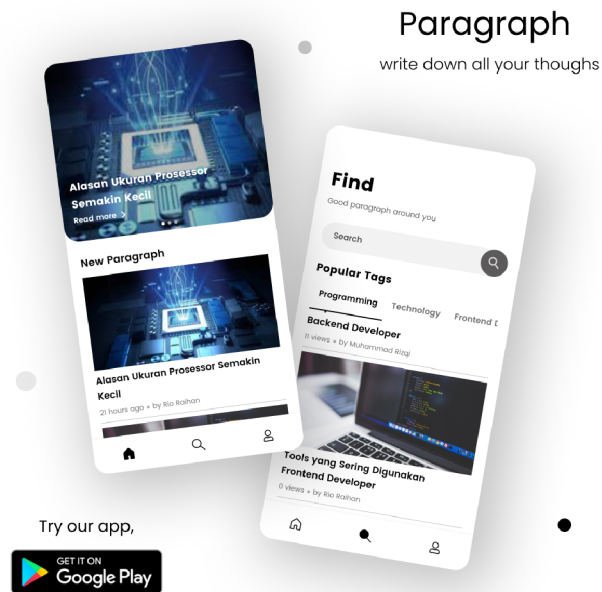
What shipped

- Short articles posted by other users
- Users can write and publish their own short articles
- Trending posts and popular tags
- Post search with multi-field filtering

LINKS

play.google.com/store/apps/details?id=com.rizqi.paragraph

github.com/m-rizqi/paragraph



Purwoasri Village Government Website

PROBLEM

The village government of Purwoasri, Pacitan had no official web presence. Public announcements, village profile, and local information had no single place to live online.

Whatever was built had to be maintainable by village staff after handover, without a developer on call.

Draft copy, written from the portfolio PDF, which carried only the live link. Send me the details and I'll replace this.

site screenshot
purwoasri-lestari.com

How I built it

1

WordPress as the base

Chose WordPress so village staff could publish and edit content themselves, with no build step and no deployment knowledge required.

2

Content structure

Set up the village profile, announcements and information sections so each type of update has an obvious place to go.

3

Handover

Handed the publishing workflow to the village so the site keeps running long after the placement ended.

STACK

WordPress

PHP

HTML

CSS

What shipped

- A public home for the village profile and official information
- Announcement and news publishing the staff run themselves
- Still online and maintained after handover

LINK

purwoasri-lestari.com

homepage screenshot
purwoasri-lestari.com

How I work

01 FRAME

Start from the real problem

Every project here began with someone's actual constraint: a store with no POS, a parent with no time, a team with no documentation.



02 ARCHITECT

Layers before features

Data, domain and presentation layers under MVVM or Clean Architecture, so business logic stays testable and the UI stays replaceable.



03 BUILD

Agile, across disciplines

Sprints and Scrum with backend, IoT and design teams, and leading them when the project calls for it, as CTO and CTEO on two PKM-K builds.



04 SHIP

Problem to Play Store

CI/CD with Jenkins and Fastlane, Crashlytics wired in early. Four apps taken from a real problem all the way to a public release.

CONTACT

If you're building something that has to ship, let's talk.

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/m-rizqi

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